Supplementary Material for "Reproducible model-based AI planning for county-scale farmland consolidation in fragmented mountain landscapes"

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S1 In-house model-free DRL baseline configuration

The Centralised PPO and Multi-Agent Reinforcement Learning (MARL) baselines reported in main-text Section 4 (and used as the fairness anchor for the contrastive-MPC results) were trained in-house under the following protocol.

Environment. Both baselines act on the same Bishan County county-level environment (`CountyLevelEnv`) used by the contrastive MPC pipeline: 13 townships, 2,600 spatial blocks aggregating 52,515 cadastral parcels. The observation is the concatenation of 2,600 blocks $\times$ 17 block-level features plus 12 global features. The action space is a categorical choice over the 2,600 blocks, with the same MaskablePPO-style action mask described in main-text Methods.

Reward. Identical to the main-text reward (Equation 1): $w_S = 4{,}000$, $w_C = 500$, $w_A = 1{,}500$, $b_B = 5$, $p_A = 2{,}000$. No method-specific re-tuning was performed, so cross-method comparisons reflect the learning procedure rather than reward engineering.

Centralised PPO. A single MaskablePPO agent with our `ParcelScoringPolicy` architecture (block-wise scoring MLP $[64, 32] \to 1$, global-feature value network). Training: 5 random seeds, 1.5×10^6 environment steps each, $\gamma = 0.995$, $\lambda = 0.95$, learning rate 1×10^{-3} linear-decayed, entropy coefficient 0.01, clip range 0.2, batch size 4,096, mini-batch size 256, 10 epochs per update. Wall-time: 8–12 GPU-hours per seed on a single A100. Cross-seed slope improvement: $-0.79\% \pm 0.36\%$, with 2/5 seeds collapsing into degenerate trajectories.

Multi-Agent Reinforcement Learning (MARL). A 13-agent variant in which each township is a separate MaskablePPO agent with shared parameters and a centralised value function over the global state, trained jointly. Same hyperparameters as Centralised PPO. 5 seeds, all stable. Cross-seed slope improvement: $-0.81\% \pm 0.10\%$ (CV $\approx 12\%$).

Why these are the relevant baselines. Both baselines are task-faithful: same environment, same action space, same reward function, same evaluation protocol (5-episode deterministic rollout) as the contrastive MPC pipeline. The wall-time and seed-failure characteristics motivate the search for a less compute-intensive learned-model alternative; the slope-improvement magnitudes (-0.8% range) provide a non-trivial reference against which the contrastive-MPC result ($-1.289\% \pm 0.079\%$ on Bishan) is then compared.

S2 Real-county initial state

The two real-county datasets anchoring the cadastral evaluation in main-text Section 2 are summarised in Table S1. All values are extracted from the Third National Land Survey records and aggregated to spatial blocks (the planning units used by the model); initial-state metrics are computed from the cadastral data before optimisation.

Table S1: Real-county initial state for Bishan District (Chongqing) and Neijiang Dongxing District (Sichuan).

	Bishan, Chongqing	Neijiang Dongxing, Sichuan
Townships	13	29
Cadastral parcels	52,515	76,376
Spatial blocks (planning units)	2,600	3,711
Initial farmland slope (deg)	9.62	10.55
Initial contiguity index	3.59	2.63
Initial *baimu fang* count	109	384
Initial *baimu fang* area (ha)	46,844	74,342

S3 Bishan replication regimes: cross-regime reconciliation

The main text cites Bishan slope-improvement numbers in three regimes (§5 "Three Bishan replication regimes"). The numbers differ along three axes—evaluation protocol, prepare pipeline, and parcel-set policy—and we tabulate them here for transparent cross-reference. All three regimes evaluate the same underlying contrastive-MPC pipeline (λ_{rank}=5.0, H=5, K=50, greedy continuation, γ=0.99); none is a re-tuning of the algorithm. The shipped ensemble used in regimes B and C is the public release `paper/checkpoints/bishan/shipped_onnx/` (3-member ensemble, λ_{rank}=5.0); the regime-A ensembles are the public release `paper/checkpoints/bishan/contrastive_5seed/` (five 3-member ensembles, seeds 0–4).

Why three regimes coexist. The research-side prepare (regime A) was used to generate the paper's headline main-text Section 4 numbers and the published λ-ablation main-text Table 1 of main-text Section 3 because it is the prepare pipeline against which the in-house DRL baselines were originally trained, ensuring an apples-to-apples comparison. The package-side prepare (regimes B and C) is the public-facing toolchain shipped with the repository; it differs from the research-side prepare in two respects relevant to the present numbers: (a) it computes block aggregation from the DLTB shapefile directly rather than from a precomputed `DLTB_with_slope.gpkg`, so the block boundaries differ slightly (2,640 vs. 2,600 blocks); (b) it preserves true non-square pixels of geographic-CRS DEMs (commit `dc3a01c`), which fixes a slope-underestimation bug in the legacy prepare and exposes a $\sim$1.2° taller mean initial slope. Regime A and regimes B/C therefore present the same algorithm operating on slightly different prepared inputs, which we treat as deployment-side variance rather than as an algorithmic effect. The 82-parcel difference between regime-B's env-side count (53,004) and regime-C's verification-side count (53,086) is auxiliary: the verification script keeps every parcel with a slope value and a farm/forest `ORIG_DLBM` regardless of township-registry membership, while the env enforces township membership; the additional 82 parcels all carry `CHG_FLAG` = 0 and contribute zero to every delta.

Table S2: The three Bishan replication regimes. Regime A is the canonical headline of main-text §5. Regime B is the toolchain replication of main-text §7. Regime C is the deployment-dataset run that anchors the GIS-grounded verification of main-text Methods *Independent GIS-grounded verification* and *Replacing the ensemble with the true environment.* The cross-regime band (−1.3 to −2.0 pp on slope) defines the deployment-side variance of the same underlying algorithm; the gap between A and B/C is morphological—the corrected geographic-CRS slope step in the package-side prepare exposes a ∼1.2° taller initial mean slope, leaving more headroom for reduction within the same 100-step episode budget.

Regime	A (lab headline)	B (toolchain)
Slope change %	-1.289 ± 0.079	-1.544 ± 0.041
ΔContiguity	$+0.0160 \pm 0.0016$	not reported in main text
ΔBaimu count	$+3.4 \pm 1.0$	not reported in main text
ΔBaimu (ha)	-312.2 ± 34.3	not reported in main text
Replication unit	5 ensembles × 1 ep	1 ensemble × 5 ep
Variance estimator	cross-seed (5 ensembles)	cross-episode (5 episodes)
Prepare pipeline	research-side (legacy)	package-side (corrected CRS
Initial slope $\bar{\sigma}_0$	8.41°	9.62°
Blocks N_{blocks}	2,600	2,640
Parcel set	52,515	53,004 (env)
Anchor location in main text	main-text Section 4 prose; SI Table S9	main-text Section 8 Methods `tool`
Source artefact	`paper/checkpoints/bishan/contrastive_5seed/`	`paper/checkpoints/bishan/shipp`

Cross-regime reproduction by the open-source toolchain. An end-to-end reproduction of all five §5 metrics on the package-side prepare from raw DLTB shapefiles plus DEM tiles (`paper/repro_artifacts/macos_2026-05-29/bishan_5seed.json`) achieves slope $-1.7384 \pm 0.024\%$, contiguity $+0.01442 \pm 0.001$, baimu count $+3.40 \pm 1.36$, baimu area -478.64 ± 35.0 ha. This is closest in magnitude to regime B/C and reproduces the qualitative cross-county finding (Bishan loses qualifying area; Dongxing gains it) end-to-end without access to the research-side prepare. A separate package-CLI 5-episode greedy run on the shipped ensemble (`runs/bishan/mpc_output/mpc_summary.json`) yields slope $-2.0392 \pm 0.0010\%$, baimu area -499.12 ± 0.17 ha, also closest in magnitude to regime C but with the ∼0.04 pp offset from the regime-C verification anchor of −2.001% that we attribute to ensemble-determinism path differences across re-runs of the deployment dataset (the shipped ensemble is deterministic across the action-selection path under fixed seed but the 5-episode greedy mean averages over five episode-0 trajectories that differ only in the episode-level random-rollout step the variance estimator consumes; §S3 Table S2 caption). The full reproduction recipe is documented in `docs/REPRODUCE.md`; the per-seed JSON outputs are released alongside the toolchain.

S4 Statistical tests reported in the main text

Table S3 consolidates the inline statistical tests cited in the main text. Tests against in-house DRL baselines (Centralised PPO, MARL) use Welch's two-sample t-test on summary statistics (the per-seed run logs of the in-house baselines are not redistributed; means and sample standard deviations are reported in §S1). Tests between contrastive (λ=5) and MSE-only (λ=0) ensembles on Buchanan are paired tests over n=5 matched ensemble training seeds, using both Welch's paired t and a Wilcoxon signed-rank test as a non-parametric robustness check, plus a TOST equivalence test at $\Delta = 0.5$ reward units ($\sim$0.22% of mean reward). All tests are two-sided unless stated otherwise. Effect sizes are Cohen's d (independent) or d_z (paired); pooled standard deviations follow Hedges & Olkin's convention.

Table S3: All inline statistical tests reported in main text §5 and §6, with raw inputs and effect sizes. Test types: W2 = Welch's two-sample t; PT = paired t; WSR = Wilcoxon signed-rank; TOST = two-one-sided-tests equivalence at the stated margin Δ. Cohen's effect-size thresholds (Cohen 1988): $|d|>0.8$ large, $|d|>1.2$ very large.

Comparison	Group 1 (mean±sd, n)	Group 2 (mean±sd, n)	Test	Statistic	p	Effect size
Bishan slope: contrastive vs Centralised PPO	−1.289±0.079% (5)	−0.79±0.36% (5)	W2	$t=-3.01$, df=4.5	0.034	$d=-1.90$ (very large)
Bishan slope: contrastive vs MARL	−1.289±0.079% (5)	−0.81±0.10% (5)	W2	$t=-8.01$, df=7.9	$< 10^{-4}$	$d=-5.07$ (very large)
Buchanan reward: contrastive (λ=5) vs MSE-only (λ=0)	229.39±0.030 (5)	229.20±0.157 (5)	PT	$t=+2.61$, df=4	0.060	$d_z=+1.17$ (very large)
Buchanan equivalence at $\Delta=\pm 0.5$ reward units	—	—	TOST	$t_{\mathrm{lower}}=+9.25$, $t_{\mathrm{upper}}=-4.04$, df=4	0.008	equivalence asserted
Buchanan reward (non-parametric robustness)	median 229.40 (5)	median 229.26 (5)	WSR	W=0.0	0.063	—

The Buchanan paired t-test result (p=0.060, marginally outside the conventional 0.05 superiority threshold) and the TOST equivalence result (p_{TOST}=0.008 at a ±0.22%-of-mean-reward margin) jointly support the main-text reading: a small, reproducible, but operationally negligible reward advantage of the contrastive variant in the regime where MSE training already acquires the ranking signal as a by-product. Whether the small superiority signal would survive at $n > 5$ is unresolved at the present sample size; the equivalence claim is the conservative reading the main text adopts. Source per-seed reward data: `runs/restoration/buchanan_va/5seed_results.json` (contrastive, λ=5) and `5seed_lam0_results.json` (MSE-only, λ=0); the `paper/checkpoints/restoration/buchanan_va/` directory holds the corresponding ensemble weights for both conditions.

S5 MPC optimisation ablation on the MSE-only ensemble (Supplementary Table S1)

Table S4 below reports the full MPC optimisation ablation cited in main-text Section 3 (paragraph beginning "Four independent operational failures"). All variants use the same MSE-only world-model ensemble (Bishan District, 5-episode evaluation) and modify a single dimension of the planner. None exceeds the random-sampling baseline; the catastrophic failure of greedy continuation under the MSE-only ensemble (slope +0.029%, wrong direction) is the variant that contrastive training subsequently rescues into the strongest variant tested (main-text Table 1, the λ=5.0 rows).

S6 Synthetic-benchmark secondary metrics (Tables S2–S4)

Table S5 reports the slope-improvement profile across the seven synthetic landscapes, summarised in main-text Section 5. The three secondary metrics referenced in main-text Section 5 (Δcontiguity, Δ*baimu fang* count, and Δ*baimu fang* area, all under the same 50-pair swap budget and n=5 seeds per cell) are tabulated below the slope profile. Together these summarise the full $5 \times 7 \times 5 = 175$-cell sweep. Aggregation script: `scripts/si_tables_v7.py` in the released code repository.

Table S4: MPC optimisation ablation across four dimensions, on the MSE-only world-model ensemble (Bishan District). All variants start from the optimal H=5, K=50 random-reward configuration and modify one dimension. None exceeds the random-sampling baseline; aggressive exploitation (greedy continuation) actively reverses the slope direction. The diagnostic shows that the failure is structural to the learned model rather than to the search procedure.

Dimension	Configuration	Slope %	Reward	Diagnosis
Continuation	Random (baseline)	-0.209 ± 0.008	−18.52	—
	Greedy	$+0.029 \pm 0.005$	−95.29	catastrophic
Scoring	Reward (baseline)	-0.209 ± 0.008	−18.52	—
	Slope-only	−0.001	−98.95	zero discrimination
Terminal value	No value (baseline)	-0.209 ± 0.008	−18.52	—
	Value-guided (v=1.0)	-0.172 ± 0.042	−27.96	no improvement
Search budget	K=50 (baseline)	-0.209 ± 0.008	−18.52	—
	K=100	-0.167 ± 0.053	−8.94	no improvement
	K=50, n_r=3	-0.171 ± 0.046	−28.33	no improvement

Table S5: Slope-improvement profile of five methods across the seven synthetic landscapes, n=5 seeds per cell (175 cells total). Values are mean ± sample standard deviation of the relative slope change in percent (negative is better). The rightmost column reports median wall-time per cell (seconds for fast methods, hours for learning-based). Best mean per row in bold. Contrastive MPC is the strongest method on the five planning-favourable presets (the two real-data clones and the three medium–large fragmented presets); the genetic algorithm wins on `plain_small_cons`, where the small action space lets a population search cover the optimum within budget. Numbers regenerated from `benchmark/results/` via `scripts/aggregate_results.py`.

Preset (n_{blocks})	Random	Greedy	GA	PPO	MPC	Wall (MPC)
`bishan_clone` (2,600)	−0.026±0.003	−0.052±0.009	−0.060±0.009	−0.074±0.015	**−0.089±0.014**	4.1h
`neijiang_clone` (3,700)	−0.011±0.002	−0.025±0.004	−0.026±0.005	−0.036±0.011	**−0.040±0.007**	6.1h
`plain_small_cons` (800)	−0.544±0.072	−0.996±0.268	**−2.757±0.203**	−2.234±0.264	−2.226±0.220	2.0h
`plain_large_cons` (4,500)	−0.071±0.015	−0.128±0.020	−0.309±0.018	−0.419±0.060	**−0.424±0.032**	10.9h
`plain_medium_frag` (2,600)	−0.132±0.028	−0.310±0.080	−0.651±0.029	−0.773±0.076	**−0.805±0.069**	6.3h
`mixed_medium_frag` (2,600)	−0.078±0.018	−0.173±0.026	−0.405±0.025	−0.414±0.076	**−0.431±0.033**	5.3h
`hilly_small_cons` (800)	−0.090±0.028	−0.152±0.012	−0.295±0.050	−0.260±0.035	**−0.271±0.047**	1.1h

Table S6: Synthetic-benchmark contiguity change (ΔContiguity, absolute units). Mean ± standard deviation across n=5 seeds. Higher (more positive) is better; per-row best in bold.

Preset (n_{blocks})	Random	Greedy	GA	PPO	MPC
`bishan_clone` (2,600)	0.0016±0.0007	0.0009±0.0006	**0.0253±0.0023**	0.0027±0.0005	0.0032±0.0009
`neijiang_clone` (3,700)	0.0014±0.0004	0.0012±0.0001	**0.0159±0.0005**	0.0015±0.0004	0.0023±0.0005
`plain_small_cons` (800)	0.0057±0.0023	0.0036±0.0010	0.0018±0.0057	**0.0069±0.0014**	0.0068±0.0015
`plain_large_cons` (4,500)	0.0010±0.0001	0.0008±0.0002	0.0010±0.0003	**0.0014±0.0002**	0.0012±0.0003
`plain_medium_frag` (2,600)	0.0018±0.0004	0.0018±0.0005	0.0005±0.0006	0.0022±0.0003	**0.0026±0.0004**
`mixed_medium_frag` (2,600)	0.0022±0.0006	0.0018±0.0003	**0.0059±0.0016**	0.0031±0.0007	0.0032±0.0008
`hilly_small_cons` (800)	0.0058±0.0020	0.0067±0.0017	**0.0734±0.0055**	0.0092±0.0020	0.0095±0.0013

Table S7: Synthetic-benchmark large-patch count change (Δ*baimu fang* count, ≥ 6.67 ha qualifying patches gained per episode). Mean $\pm$ standard deviation across n=5 seeds. Higher is better; per-row best in bold.

Preset (n_{blocks})	Random	Greedy	GA	PPO	MPC
bishan_clone (2,600)	0.0±2.2	−0.2±2.6	−3.8±4.4	**9.4±2.6**	9.2±2.3
neijiang_clone (3,700)	0.6±0.9	−1.4±1.5	−4.4±3.2	**6.4±2.1**	4.6±3.4
plain_small_cons (800)	0.0±0.0	0.2±1.1	**0.8±2.0**	−0.4±2.5	0.4±2.1
plain_large_cons (4,500)	0.0±0.0	0.0±0.0	0.2±0.4	**0.8±0.8**	0.0±0.7
plain_medium_frag (2,600)	0.6±0.9	−0.6±1.3	1.4±1.8	**1.6±1.1**	0.4±1.3
mixed_medium_frag (2,600)	−0.2±1.3	0.0±1.2	**1.2±2.2**	−1.0±2.2	−2.4±1.1
hilly_small_cons (800)	−1.0±0.7	−0.4±0.9	−3.2±2.2	3.4±1.7	**9.8±1.5**

Table S8: Synthetic-benchmark large-patch area change (Δ*baimu fang* area, ha). Mean $\pm$ standard deviation across n=5 seeds. The sign of this metric is morphology-dependent (see main-text Section 5 discussion of the Bishan vs. Neijiang reversal); per-row best (most positive) in bold for reference.

Preset (n_{blocks})	Random	Greedy	GA	PPO	MPC
bishan_clone (2,600)	9.8±11.1	11.9±15.0	155.3±40.3	134.7±26.1	**177.5±31.7**
neijiang_clone (3,700)	15.2±9.2	27.8±16.4	**190.8±33.2**	122.4±44.8	167.6±23.0
plain_small_cons (800)	20.2±8.9	14.7±16.2	11.5±19.5	19.0±25.3	**30.5±17.5**
plain_large_cons (4,500)	9.2±8.9	39.7±13.3	12.8±13.4	40.1±55.3	**40.3±15.3**
plain_medium_frag (2,600)	22.8±10.5	38.5±22.8	34.2±34.3	**54.0±40.1**	52.8±26.8
mixed_medium_frag (2,600)	14.2±7.3	28.4±13.6	60.7±24.8	**106.8±24.2**	102.1±13.1
hilly_small_cons (800)	21.6±14.2	4.6±13.9	**167.7±41.2**	100.3±30.3	131.9±43.5

S7 Per-seed Bishan multi-objective profile

Main-text Section 4 reports the cross-seed mean of all four contrastive-MPC primary objectives on Bishan (λ_{rank}=5.0, greedy, H=5, K=50). The per-seed values that produce that mean are listed in Table S9. Initial state: slope 9.616, contiguity 3.585, 109 *baimu fang* totalling 46,844 ha. Slope and contiguity changes are both directionally correct on every seed; the count of large patches ($\geq$6.67 ha) increases by +3.4 on average; total qualifying area decreases by ~312 ha because the agent consolidates many small patches into a smaller number of larger, contiguous patches—the deliberate trade-off discussed in main-text Discussion.

Table S9: Per-seed multi-objective profile of contrastive MPC on Bishan across 5 independently trained ensembles (1 episode each).

Seed	Slope %	ΔContiguity	ΔBaimu #	ΔBaimu (ha)
0	−1.4235	+0.0150	+3	−324.9
1	−1.2668	+0.0165	+3	−283.6
2	−1.3250	+0.0159	+2	−372.7
3	−1.2264	+0.0139	+5	−278.2
4	−1.2018	+0.0186	+4	−301.9
Mean	$\mathbf{-1.289 \pm 0.079}$	$\mathbf{+0.0160 \pm 0.0016}$	$\mathbf{+3.4 \pm 1.0}$	$\mathbf{-312.2 \pm 34.3}$

S8 Toolchain pre-flight checks

Tool 5 of the ArcGIS Pro toolchain (main-text Section 8) validates the user environment before any planning step is allowed to run. The complete list of checks is:

1. **Python interpreter**: ArcGIS Pro 3.6+ `arcgispro-py3` or compatible Python 3.10+ environment.
2. **Spatial Analyst extension**: licence checked out (required by Tool 1 for slope derivation).
3. **Required packages**: `gymnasium` ($\geq$ 0.29), `libpysal` ($\geq$ 4.7), `onnxruntime` ($\geq$ 1.16), `numpy`, `pandas`, `scipy`.
4. **ONNX-ensemble shape compatibility**: if a pre-trained ensemble is provided, the baked-in N_{blocks} matches the prepared dataset.
5. **Coordinate reference system**: the input shapefile's CRS resolves to a metric projection (default `EPSG:32648`; user-overridable).
6. **Disk space**: at least 5 GB free in the prepared-data directory.

Tool 5 prints an `[OK]` or `[FAIL]` verdict for each item and refuses to proceed with downstream tools if any required item fails. The 70-second smoke test on the bundled 2-township subset of Neijiang Dongxing District exercises every check end to end and is the recommended onboarding action for new users of the toolchain.

S9 Data and code release

The synthetic benchmark, the pre-trained Bishan ensemble, the ArcGIS Pro toolbox source, and the aggregation/sweep scripts referenced throughout this paper are released under the licences indicated in the main-text Data and Code Availability statements. Repository structure at the public release:

- `LandUseOptimization_P9.pyt`: ArcGIS Pro Python Toolbox manifest.
- `farmland_mpc/`: pure-Python core (loss functions, ensemble training, MPC planner, ONNX export).
- `benchmark/`: synthetic generator, presets, baseline runners, sweep manifest, and the 175-cell results.
- `docs/BENCHMARK_RESULTS.md`: per-(preset, method) aggregated results derived from `benchmark/results/`.
- `scripts/aggregate_results.py`, `scripts/si_tables_v7.py`: aggregation and SI-table generators.
- `environment.yml`, Colab demo notebook, integration smoke test.

S10 Verification stack: full per-episode data (Tables S5–S8)

The main-text Methods sections (`Independent GIS-grounded verification`, `Direct measurement of ensemble 1-step prediction accuracy`, and `Replacing the ensemble with the true environment`) summarise four of the five independent verification layers underwriting the Bishan headline result (the fifth, Neijiang cross-county replication, is in main-text Section 4). Tables S10–S13 give the per-replicate, per-channel numbers for each verification layer so external readers can audit them without rerunning the released scripts.

Layer (i): GIS-grounded recomputation. Table S10 compares the planner-reported deltas (from `mpc_summary.json`) against an independent recomputation that reads only the optimised DLTB shapefile, joins per-parcel slope from the prepared DEM raster, builds Queen contiguity via `libpysal`, and applies the same metric definitions as the main-text Methods. The recomputation script (`validate_optimized_shp.py`) shares no source with the optimisation pipeline.

Table S10: Independent GIS recomputation versus planner-reported deltas, Bishan run_1, 53,086 swappable parcels. The recomputation invokes no learned model. Differences are at floating-point reduction noise level.

Quantity	Recomputed (independent)	Planner-reported	$\lvert\Delta\rvert$
Δ slope (%)	-2.0000	-2.0006	6×10^{-4} pp
Δ contiguity	$+0.012739$	$+0.012748$	8×10^{-6}
Δ baimu count	$+8$	$+8$	0 (exact)
Δ baimu area (ha)	-473.2950	-473.2950	$< 10^{-4}$ ha
farm→forest swaps	428	428	0 (exact)
forest→farm swaps	428	428	0 (exact)

Layer (ii): ensemble member-subset ablation. Table S11 reports the three replicates that constitute the variance estimator used in the main-text Methods caveat on five-seed determinism. Each replicate runs the full MPC pipeline (H=5, K=50, greedy continuation, scoring=reward) with a different 2-of-3 ensemble member subset. The full ensemble's result lies inside the mean ± sd of the subset distribution on every reported metric.

Table S11: Per-replicate results for the member-subset ablation. The three replicates exhaust C(3,2)=3 distinct subsets. Full-ensemble values reproduced from main-text Table 4 / `mpc_summary.json` for comparison; they fall inside the subset mean±sd on every metric.

Replicate (members)	Δ slope (%)	Δ cont	Δ baimu count	Δ baimu (ha)
{0, 1}	−2.0222	+0.01361	+5	−475.02
{0, 2}	−1.9706	+0.01437	+5	−443.12
{1, 2}	−2.0024	+0.01128	+6	−505.39
Mean ± sd (n=3)	-1.9984 ± 0.0260	$+0.01309 \pm 0.00161$	$+5.33 \pm 0.58$	-474.51 ± 31.14
Full ensemble {0,1,2}	−2.0006	+0.01275	+8	−473.30

Layer (iii): direct ensemble 1-step accuracy. Table S12 reports the per-channel prediction error of the deployed 3-member ensemble against the true environment, accumulated over the 100 in-distribution steps that reproduce episode 0 of `mpc_summary.json`. The reward head is strongly under-fit but Spearman rank correlation between predicted and true reward is positive, and slope-channel error is sub-variance.

Table S12: Ensemble 1-step prediction error over the same 100-step trajectory MPC visits (Bishan, seed 0). True-reward statistics are computed from the actual env.step return. Spearman rank correlation between predicted and true reward is +0.216. Pearson correlation between ensemble disagreement r_{std} and absolute prediction error is +0.077, i.e. disagreement does not flag where members are wrong.

Quantity	Value
Reward MAE	0.818
Reward RMSE	1.540
Reward median |err|	0.170
Reward p90 |err|	3.148
Reward p99 |err|	5.075
r_{true} range / std	$[-3.04, +6.36]$ / 1.397
r_{pred} range / std	$[+0.43, +0.67]$ / 0.069
Naive constant-mean baseline MAE	0.828
Spearman $(r_{\text{pred}}, r_{\text{true}})$	+0.216
Ensemble r_{std} mean / median / max	0.027 / 0.030 / 0.075
Pearson $(r_{\text{std}}, \lvert\text{err}\rvert)$	+0.077
$gf[4]$ slope-improvement MAE	0.00238 (channel std 0.00521)
$gf[6]$ baimu-count-norm MAE	0.00270 (channel std 0.00817)

Layer (iv): true-environment MPC counter-factual. Table S13 compares the ensemble-MPC headline trajectory against an ablation that replaces every ensemble call with the actual env.step, under matched H/K/γ and a single seed. Implementation concessions (random rather than greedy continuation; stage-1 subsample of 200 of ∼2,000 valid actions) are forced by the cost of incremental simulation. Despite using a perfect dynamics model, the resulting trajectory is strictly worse on all three objectives.

Table S13: True-environment MPC vs. deployed ensemble MPC, Bishan, 100 outer steps, seed 0. "True-env" replaces every ensemble call with actual env.step; otherwise H=5, K=50, γ=0.99 unchanged. Implementation concessions stated in caption: random (rather than greedy) continuation, stage-1 subsample of 200 valid actions per step.

Quantity	Ensemble MPC (run_1)	True-env MPC	Δ (true − ensemble)
Δ slope (%)	−2.0006	−1.8187	+0.182 pp (worse)
Δ contiguity	+0.01275	+0.00865	−0.0041 (worse)
Δ baimu count	+8	+2	−6 (worse)
Δ baimu area (ha)	−473.30	−565.10	−91.8 ha more loss
Wall time (excl. env build)	1,448 s	358 s	true-env 4× faster
Stage-1 coverage	all valid actions (batched)	200-action random subset	—
Stage-2 continuation	greedy	random	—

Per-step trajectory matching between layer (iii) and the deployed pipeline is byte-identical at every monitored step (10, 20, . . . , 100) of episode 0; this is the most stringent reproducibility check the verification stack supports and is therefore the canonical evidence that the deployed ensemble's planning is deterministic given a fixed prepared dataset and config.